Smart Energy Consumption Analysis and Prediction using Machine Learning with Device-Level Insights

Infosys Springboard Internship Project

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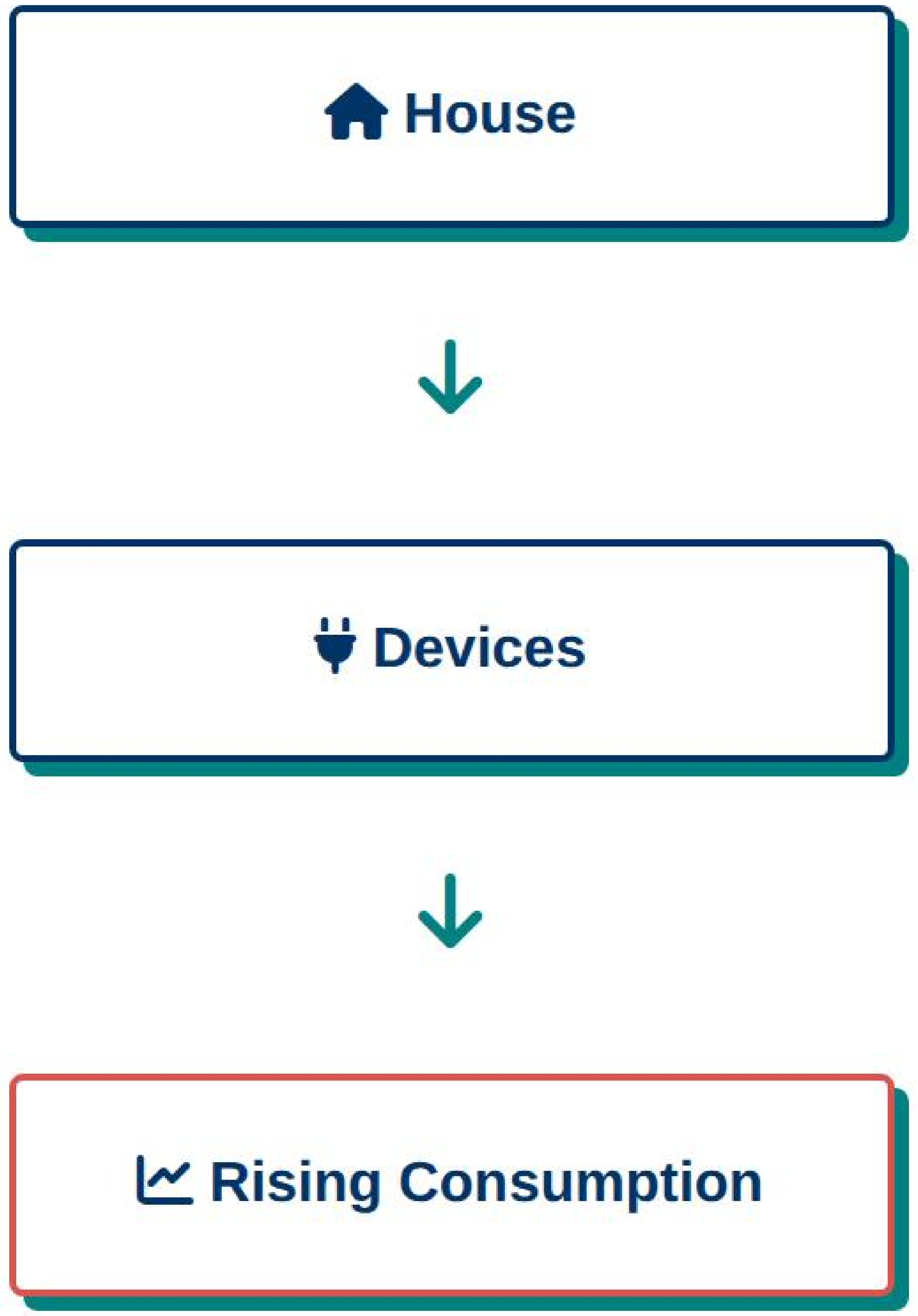
Machine Learning | Data Engineering | Deployment

PROBLEM STATEMENT

- Rising household energy consumption
- Lack of device-level visibility
- No predictive intelligence for consumption planning
- Users cannot proactively optimize electricity usage



PROJECT OBJECTIVES

- Analyze household energy consumption patterns
- Engineer time-series features for better predictions
- Compare traditional ML vs Deep Learning models
- Build a deployable prediction API (Flask)
- Create an interactive dashboard for end users (Streamlit)

DATASET DESCRIPTION

Source: Individual Household Electric Power Consumption Dataset

Scale: ~2M+ records | 9 attributes | Time-series sampling

Feature	Description
Global_active_power	Total household active power consumption (kW)
Voltage	Average voltage levels (V)
Sub_metering_1	Kitchen (dishwasher, microwave, oven)
Sub_metering_2	Laundry room (washing machine, dryer)
Sub_metering_3	HVAC system (water-heater, air-conditioner)

DATA CLEANING & PREPROCESSING

- Handled missing values (imputation via mean)
- Converted Date & Time into datetime format
- Resampled data (Hourly/Daily frequency)
- Removed anomalies and outliers
- Normalized features using MinMaxScaler for LSTM

Processing Pipeline

Raw Data

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Cleaning / Imputation

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Resampling (Hourly)

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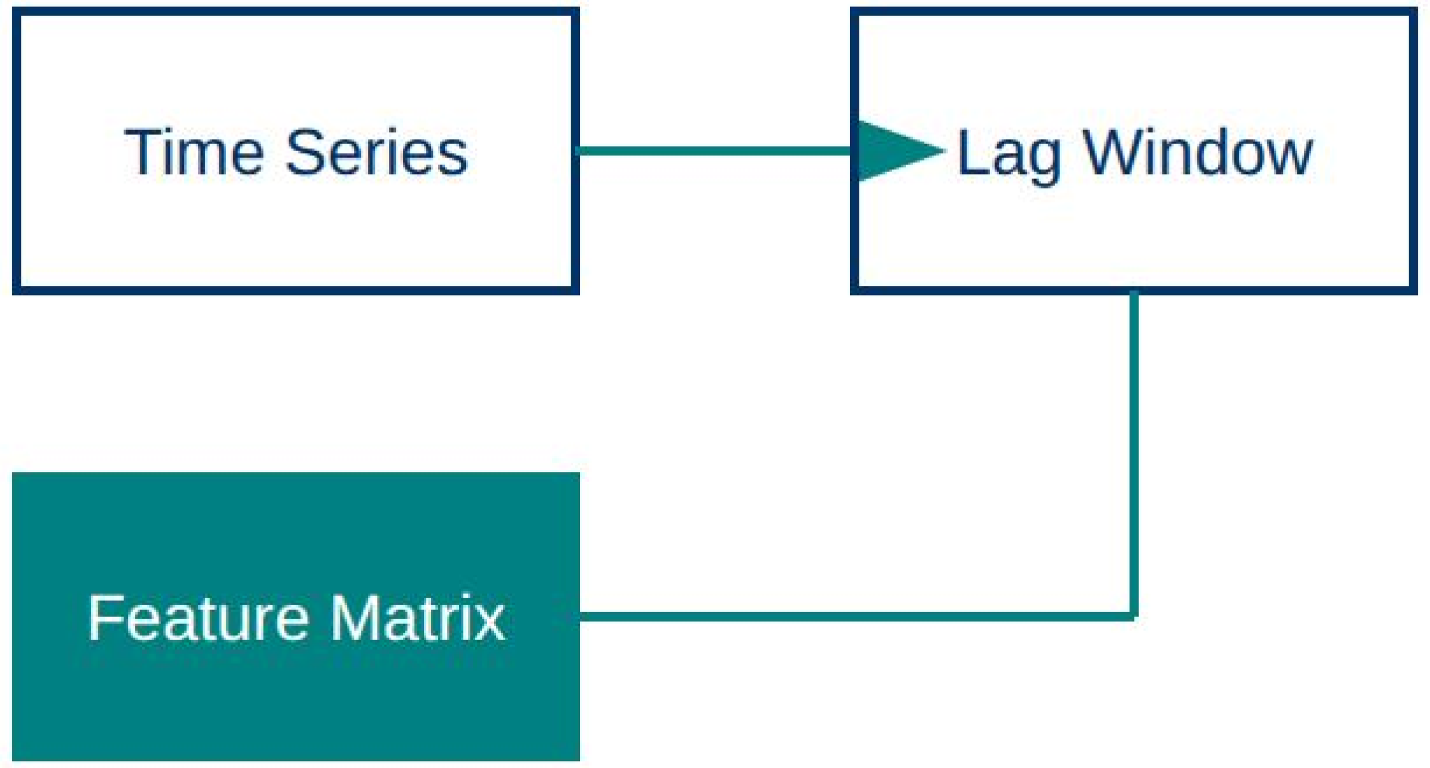
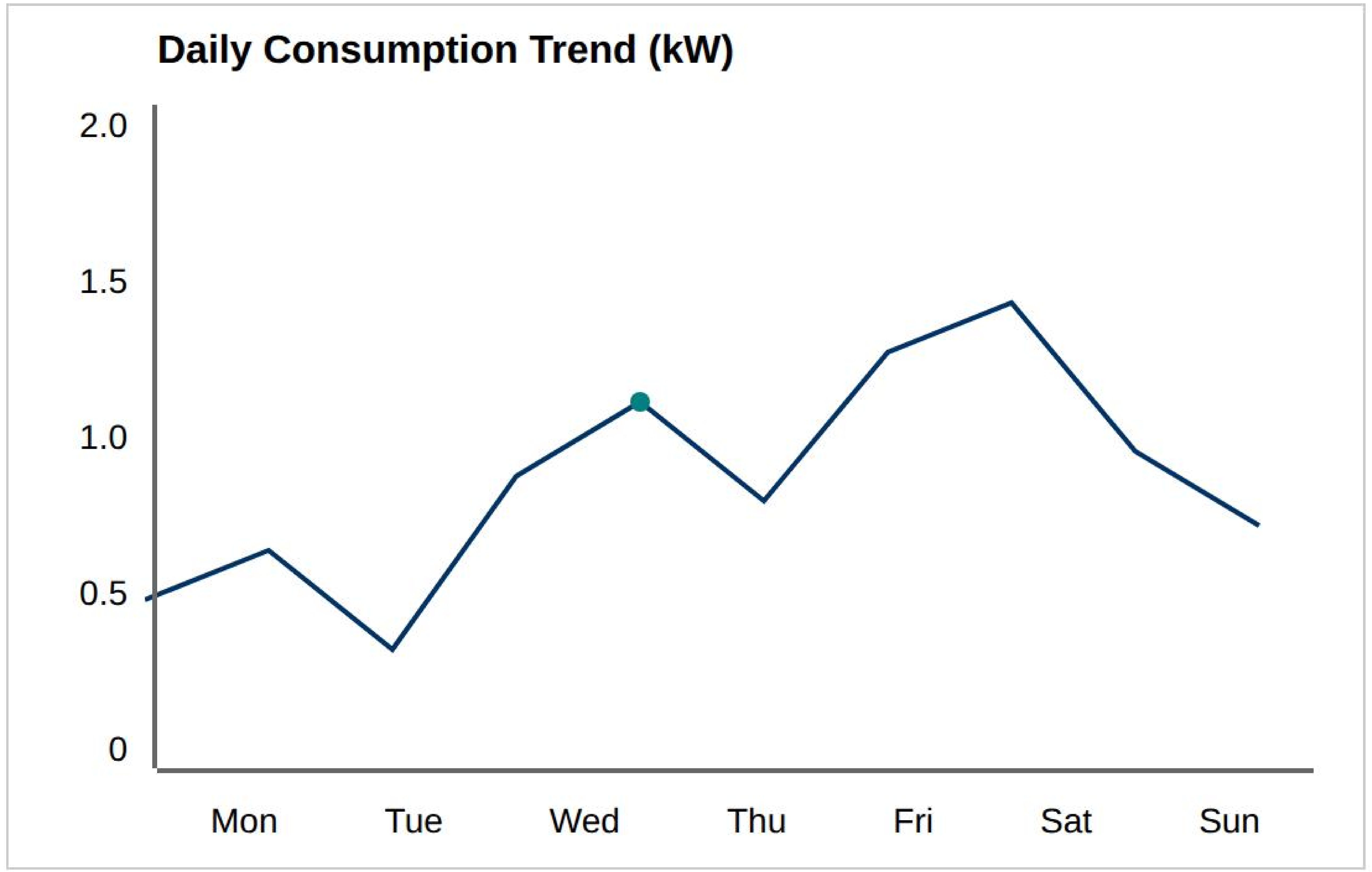
Feature Ready Data

FEATURE ENGINEERING STRATEGY

- **Lag Features:** t-1, t-2, t-24 to capture history
- **Rolling Mean:** 24-hour moving average
- **Temporal Extraction:** Hour, Day, Month extraction
- **Device-level aggregation:** Summarizing sub-meters

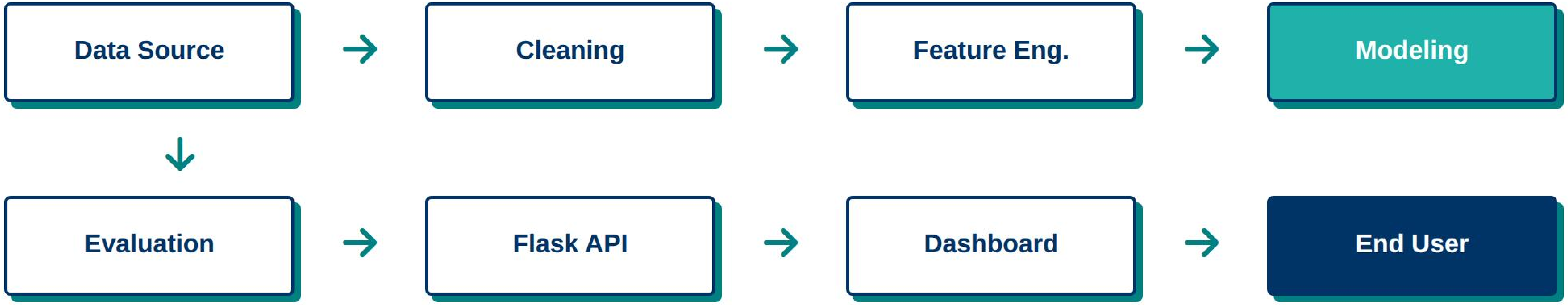


EXPLORATORY DATA ANALYSIS



- Peak usage observed during evening hours (18:00 - 21:00)
- Strong correlation between Global Active Power and Sub-metering 3 (HVAC)
- Significant seasonal variation identified across winter months

END-TO-END ML PIPELINE



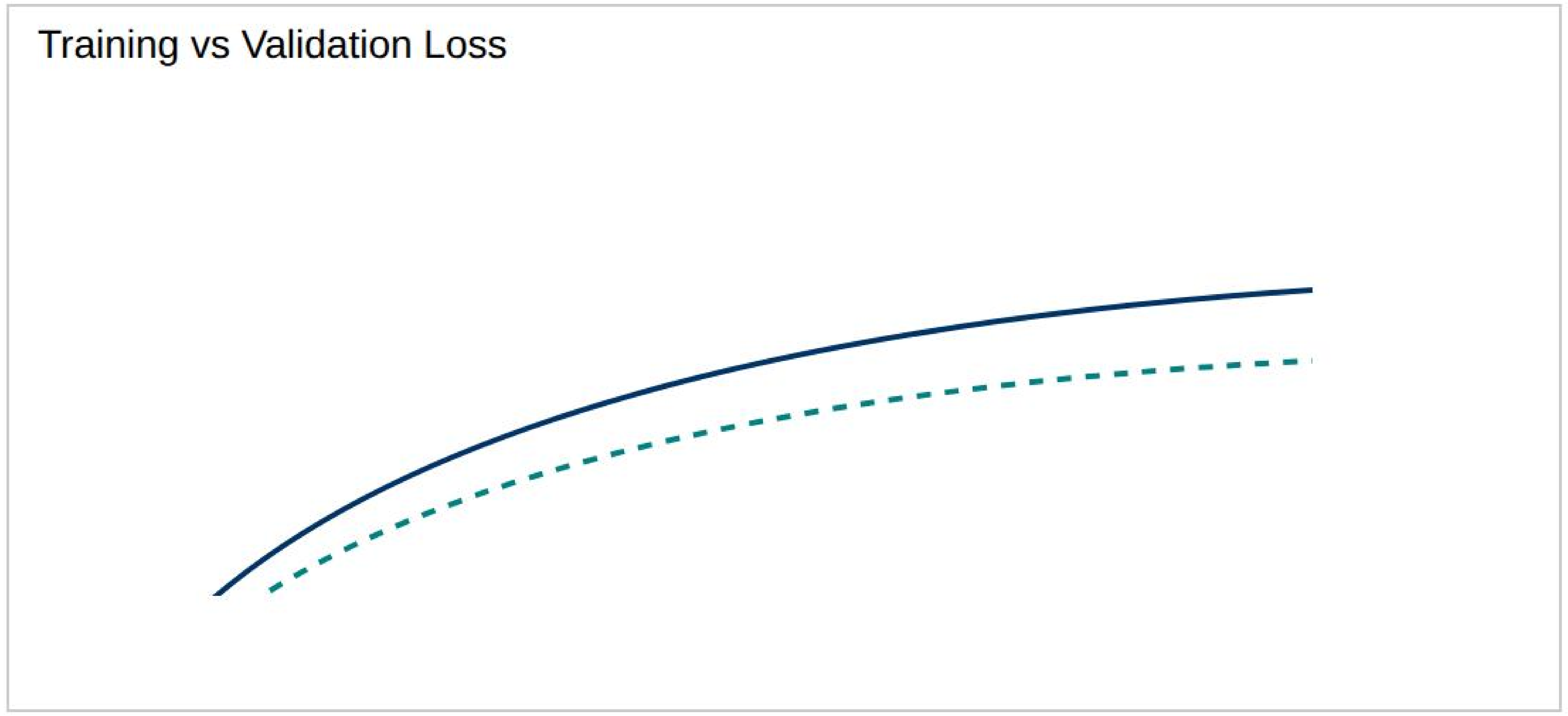
BASELINE MODEL – LINEAR REGRESSION

- Standard regression benchmark
- Extremely fast training and low resource usage
- High interpretability of feature coefficients
- Provides baseline for error reduction

Metric	Value
MAE	0.528
RMSE	0.642
R^2 Score	0.621

LSTM MODEL ARCHITECTURE

- Sequential time-series modeling
- Captures long-term temporal dependencies
- Architecture: Input → LSTM Layer (50 units) → Dropout → Dense Output



Deep Learning Architecture

Input Tensor [Lag=24]

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LSTM Layer (50 Units)

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Dense Layer (ReLU)

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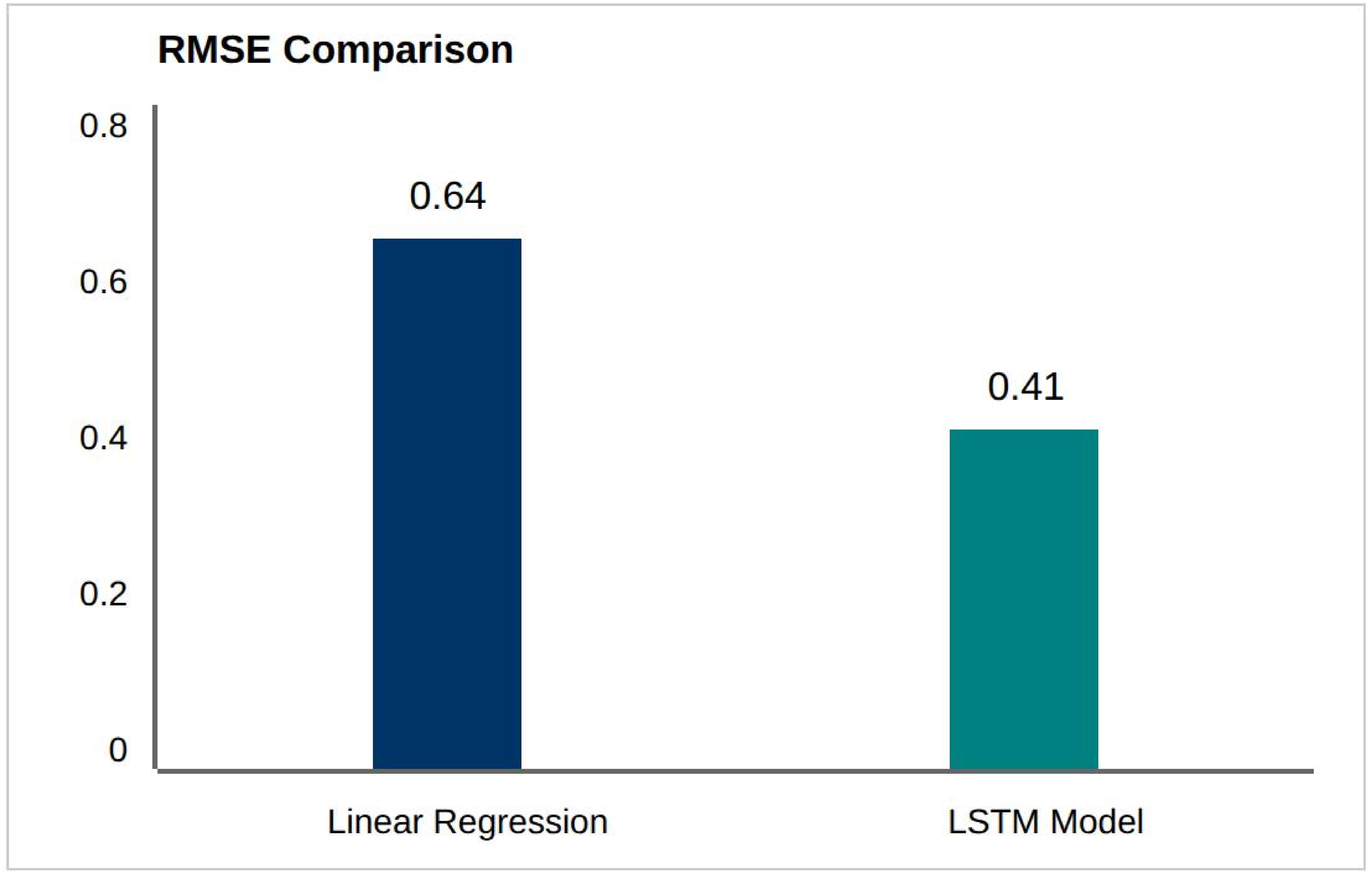
Global Active Power Prediction

MODEL PERFORMANCE COMPARISON

Model	MAE	RMSE	R^2
Linear Regression	0.52	0.64	0.62
LSTM (Proposed)	**0.34**	**0.41**	**0.79**

"LSTM reduced prediction error significantly (approx. 35% improvement in RMSE) compared to baseline."



FLASK PREDICTION API

- RESTful API built using Python Flask framework
- Accepts JSON payloads containing lag features
- Loads pre-trained LSTM model weights (.h5)
- Scalable and ready for cloud integration

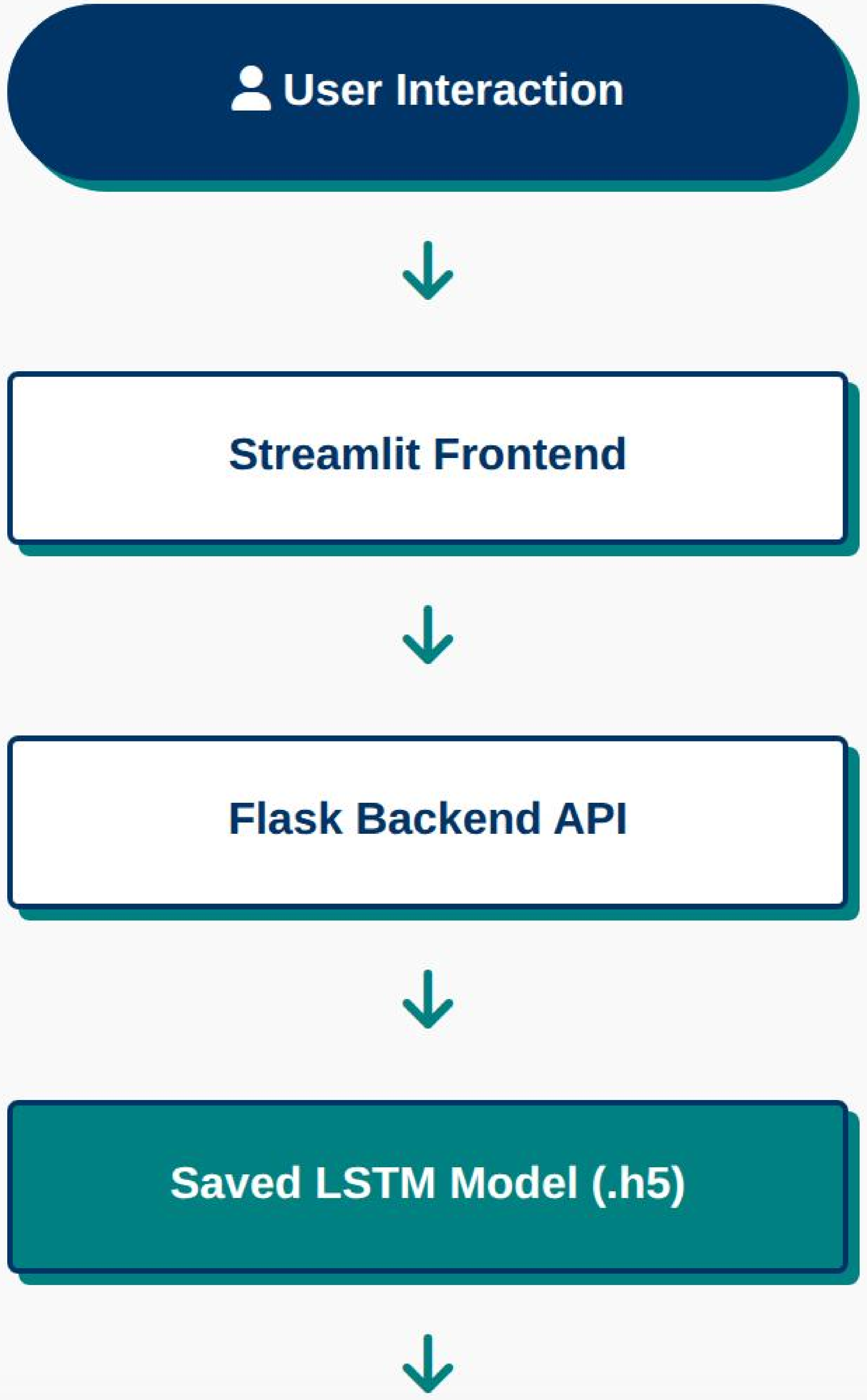
```
// Sample API Request

{ "input": [1.4, 1.2, 0.9, ...], "window": 24 }

// Sample JSON Response

{ "prediction": 1.25, "unit": "kW", "status":
"success" }
```

Streamlit UI ↔ Flask API ↔ H5 Model File

INTERACTIVE DASHBOARD

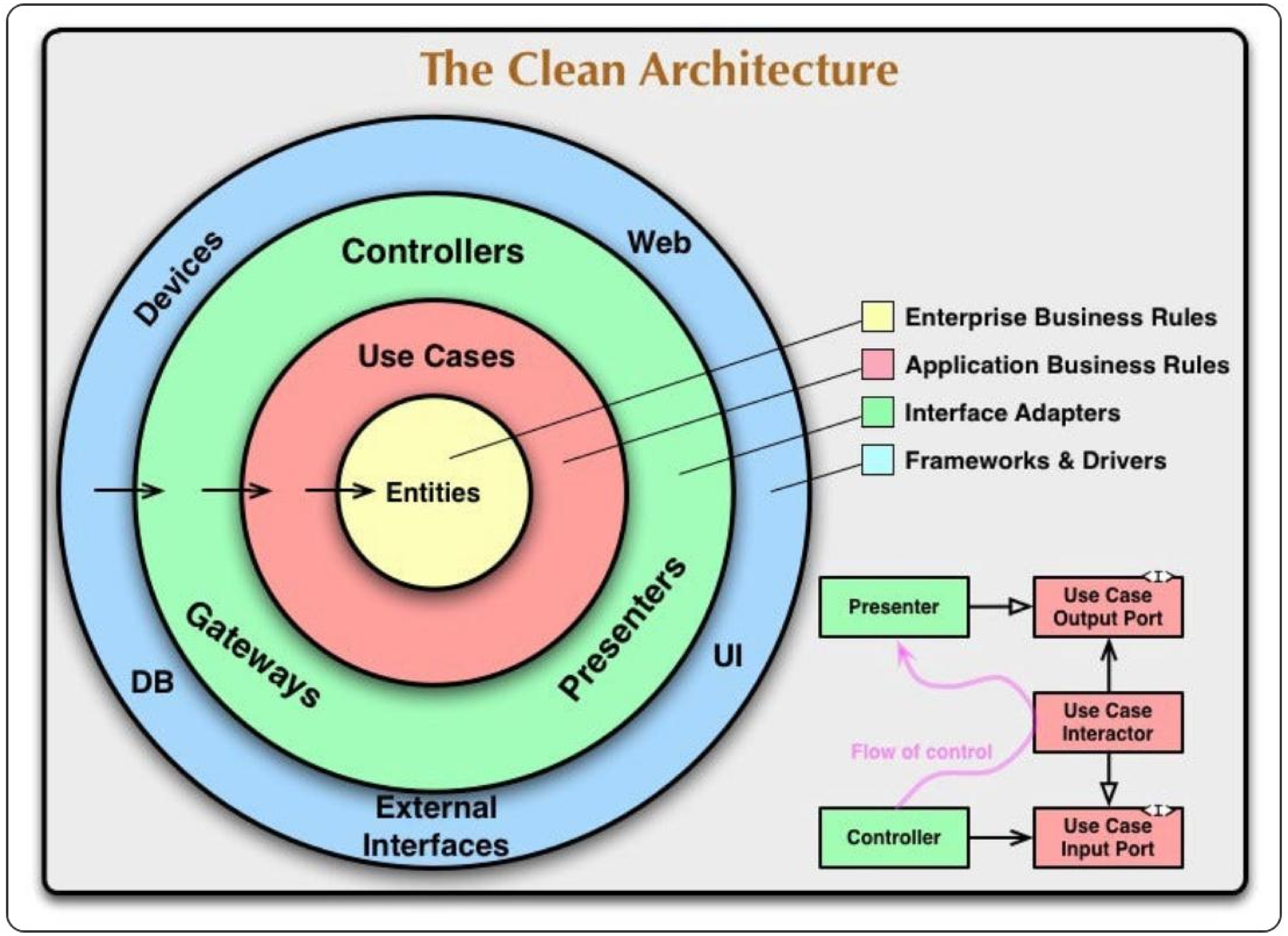
- Real-time data visualization via Streamlit
- Device-level selection for deep analysis
- Hourly, Daily, and Weekly usage trends
- Smart suggestions based on peak prediction

SYSTEM ARCHITECTURE OVERVIEW



DEPLOYMENT-READY STRUCTURE

- Modular code structure for scalability
- Decoupled Frontend (UI) and Backend (API)
- Environment management via requirements.txt
- Version control implementation (GitHub)



PROJECT IMPACT

- **Proactive Management:** Enables consumers to identify high-drain periods.
- **Cost Efficiency:** Reduces electricity planning uncertainty.
- **Transparency:** Provides granular device-level consumption data.
- **Scalability:** Flexible architecture ready for IoT-based smart home integration.

FUTURE IMPROVEMENTS

- Real-time IoT sensor integration (Edge computing)
- Full cloud deployment (AWS/Azure/GCP)
- Mobile application integration for remote monitoring
- Transformer-based forecasting for improved long-term accuracy
- Automated anomaly detection and user alerts

CONCLUSION

This project successfully demonstrates an industry-standard workflow including:

End-to-End ML Development

LSTM Time-Series Modeling

REST API & UI Integration

Industry-Ready Architecture

"Bridging Machine Learning with Real-World Energy Intelligence."

IMAGE SOURCES

https://media.istockphoto.com/id/1456630181/vector/simple-set-of-outline-icons-about-energy-efficiency-and-saving-sustainable-development.jpg?s=612x612&w=0&k=20&c=qmUSQvV98dXDSgYlvFA3Gl95nTuL_4t6k9pCmYlPfyw=

Source: www.istockphoto.com

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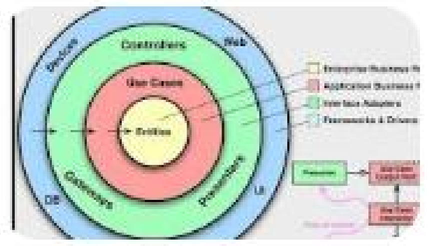

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